DO NOT FABRICATE / DO NOT ENERGIZE P48V
8 electrically designed OAM seats (first stuffing: 2 modules / 6 DNP OK – no respin).
16x Molex 218910–1115, 5.00 mm stack. 12–layer 2.0 mm stuffed–switch TARGET. Outline 492 x 372 mm.
4x2 of 103x166 mm KOZ = 412x332 mm INFERRED tiling (not a UBB drawing).
All 8: named PE toward 2x PM8536B–FEI DNP PRIMARY (x8/GCD; SW0=0–3, SW1=4–7). No CEM invented.
Host stub: X11DPH–T 3x Gen3 x16 + 4x Gen3 x8 keepout. Two CPU x16 = switch uplinks.
X11DPH–T / NH–D9 DX–3647 / B550M are NOT on this PCB. Chamber A mATX is not this PCB.
S1–S7 NOT routed (no xGMI). TEST*/RFU/DO_NOT_USE unmapped. PVREF never driven.
P48V: SB175 long–edge entry + ~15A fuse/seat + LOCAL pours on 16 Conn0 pads. NOT a 100A flood.
Molex 2026–08–18: CSA 60V (COFC 80170713) at OCP P48V; skip pins = published map, no extra NC.
OPEN: 1.2 A/contact at 48–59.5 V (2 oz). HOST_PWRGD is ENABLE. No GPU VRM. P12V2 Unknown/may be NC.
Dell D3000E–S1 is 12 V CRPS – do not mix into P48V. P3V3 = Conn0 C1/C2 only. Never r2.0.

00_DO_NOT_FABRICATE
File: sheets/00_do_not_fabricate.kicad_sch

01_Power_Clock_Reset
File: sheets/01_power_clock_reset.kicad_sch

02_Host_PCIe_Named_8Seat
File: sheets/02_host_pcie_stub.kicad_sch

03_OAM0_Connectors
File: sheets/03_oam0_connectors.kicad_sch

04_OAM1_Connectors
File: sheets/04_oam1_connectors.kicad_sch

05_OAM2_Connectors
File: sheets/05_oam2_connectors.kicad_sch

06_OAM3_Connectors
File: sheets/06_oam3_connectors.kicad_sch

07_OAM4_Connectors
File: sheets/07_oam4_connectors.kicad_sch

08_OAM5_Connectors
File: sheets/08_oam5_connectors.kicad_sch

09_OAM6_Connectors
File: sheets/09_oam6_connectors.kicad_sch

10_OAM7_Connectors
File: sheets/10_oam7_connectors.kicad_sch

11_PM8536_DNP
File: sheets/11_pcie_switch_unknown.kicad_sch

12_Unmapped_AMD
File: sheets/12_unmapped_amd.kicad_sch

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed–switch TARGET. 2x PM8536 DNP. Star–fed P
Pin names: OAM Pin map rev 1.0.xlsx – OCP generic, NOT AMD overlay
8 OAM seats. Host X11DPH–T is NOT on this PCB. No CEM cable. No 20–layer UBB.
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026–08–18; 1.2A/contact OPEN.

Sheet: /
File: MI250X_8OAM_PCBWay_Chassis_Stub.kicad_sch

Title: Rev3 8–seat PCBWay named–net chassis (NOT fab–ready)

Size: A3	Date: 2026–08–22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 1/14

GATE LIST — all must be closed before fabrication or applying power to an MI250X.

1. 48V source identified AND harnessed to the SB175. Dell D3000E-S1 is 12 V CRPS — do not tie OAM P48V to it. No PSU shopping on this PCB BOM.
2. Molex 218910-1115 voltage/skip-pin: WRITTEN YES 2026-08-18 (Brian Park / ticket 167157). CSA 60 V (COFC 80170713) at OCP P48V. Published OCP P48V map already satisfies Skip Pins — do NOT add extra NC pads. Eli ack 2026-08-19.
   STILL OPEN: 1.2 A per used power contact at 48-59.5 V (2 oz), and skip/void is NC on the same MPN. DO NOT ENERGIZE until that current follow-up is written.
3. Connector gender/stack: hermaphroditic 2189101115 mates with itself, 5 mm stack. Confirm PIN A3 orientation on a plot before tape-out (rotation 180 Inferred).
4. AMD overlay unknowns (TEST*, dual-GCD PE vs SERDES_7, SMBus map, P12V2 need, xGMI S1-S7).
5. Host X11DPH-T is NOT on this PCB. Cooler on the host is NH-D9 DX-3647 (NOT U14S). Chamber A mATX B550M 244x244 is NOT on this PCB.
6. PCIe: all 8 seats named toward 2x PM8536B-FEI DNP PRIMARY (x8 per GCD). SW0 seats 0-3, SW1 seats 4-7. Two CPU x16 uplinks. PEX8780 is docs-only cheaper 80-lane alt. No CEM cable invented. Host MPN Unknown (silk keepout only).
7. HOST_PWRGD is ENABLE. Sequencing vs P48V/P12V1/P3V3 — OCP + AMD overlay only.
8. Do not reuse the old MFC qty-5 cart for 220x120 mm. That quote is UNRELATED.
9. Per-seat ~15 A fuse MPN still Unknown (keepout only). P12V1 <=50 W and P3V3 <=5 W sources are keepouts — no GPU multiphase VRM on this PCB.
10. Do not upload to PCBWay. 12L 2.0 mm is the stuffed-switch target (plan 12-16L, not 4); 8L 2.0 mm is a cheaper DNP-switch option only.

Until then this KiCad tree is a mapping artifact. Do not send to PCBWay. DO NOT ENERGIZE P48V.

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4

Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay

Documentation sheet — no invented nets

DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /00_DO_NOT_FABRICATE/

File: 00_do_not_fabricate.kicad_sch

Title: DO NOT FABRICATE

Size: A3	Date: 2026-08-22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 2/14

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed−switch TARGET. 2x PM8536 DNP. Star−fed P4

Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay

Documentation sheet — no invented nets

DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026−08−18; 1.2A/contact OPEN.

Sheet: /01_Power_Clock_Reset/

File: 01_power_clock_reset.kicad_sch

Title: Power / clock / reset stub

Size: A3	Date: 2026−08−22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 3/14

PURPOSE: Named v1.0 PCIE_TXnP/N and PCIE_RXnP/N (n=0..15) on ALL 8 seats.

8 electrically designed seats. First stuffing may populate 2 modules / 6 DNP –
the PCB does not require a respin for the other six.

OCP pin list (module POV):
- PETp/n = module TX, host RX. AC caps on motherboard/carrier – not placed.
- PERp/n = module RX, host TX. AC caps on motherboard/carrier – not placed.

This sheet does NOT map those pairs onto a CEM x16 connector pinout.
Do NOT invent a CEM cable, SlimSAS, MCIO, or retimer BOM.

Intended 8x topology (PRIMARY DNP until stuffed):
- Two Microchip PM8536B-FEI (Switchtec PFX 96-lane Gen3, 1311-ball 37.5 mm FCBGA, 1.0 mm).
  ~$460-475, ~18 wk. PCBWay can assemble 1.0 mm without HDI; plan 12-16 layers (not 4).
  SW0 = seats 0-3 (16 US + 64 DS). SW1 = seats 4-7. x8 per GCD.
- Each MI250X GCD has its own PCIe Gen4 x16 (AMD + Hot Chips 34). Host trains Gen3.
- OAM v1.5: one PE x16 on Conn0; a second host x16 may be SERDES_7 on Conn1.
  GCD1 = SERDES_7 is Inferred – there is NO public AMD overlay proving it.
  Do not route S1-S7 as if that were proven. No xGMI.
- Each seat's 16 named PE lanes = 2x x8 (GCD0/GCD1 split is planning; PE_BIF AMD-unknown).
- Host X11DPH-T (NOT on this PCB): 3x Gen3 x16 + 4x Gen3 x8 = 80 Gen3 lanes.
  Two CPU x16 = switch uplinks (silk + connector keepout, MPN Unknown).
  Remaining 1x x16 + 4x x8 unused on this chassis.

8 x 2 x x16 = 256 downstream vs 80 host Gen3 – do not attempt x16-per-GCD.
PEX8780-AB80BI G (80-lane 35 mm) is a cheaper alt in docs only – not placed.
PM8533B-F3EI (48-lane 27 mm) is a 2-seat alt in docs only – not placed.

X11DPH-T is NOT a part on this PCB. Chamber A B550M is NOT this PCB.

◇OAM0_PCIE_TX0P
◇OAM0_PCIE_RX0P
◇OAM0_PCIE_TX0N
◇OAM0_PCIE_RX0N
◇OAM0_PCIE_TX1P
◇OAM0_PCIE_RX1P
◇OAM0_PCIE_TX1N
◇OAM0_PCIE_RX1N
◇OAM0_PCIE_TX2P
◇OAM0_PCIE_RX2P
◇OAM0_PCIE_TX2N
◇OAM0_PCIE_RX2N
◇OAM0_PCIE_TX3P
◇OAM0_PCIE_RX3P
◇OAM0_PCIE_TX3N
◇OAM0_PCIE_RX3N
◇OAM0_PCIE_TX4P
◇OAM0_PCIE_RX4P
◇OAM0_PCIE_TX4N
◇OAM0_PCIE_RX4N
◇OAM0_PCIE_TX5P
◇OAM0_PCIE_RX5P
◇OAM0_PCIE_TX5N
◇OAM0_PCIE_RX5N
◇OAM0_PCIE_TX6P
◇OAM0_PCIE_RX6P
◇OAM0_PCIE_TX6N
◇OAM0_PCIE_RX6N
◇OAM0_PCIE_TX7P
◇OAM0_PCIE_RX7P
◇OAM0_PCIE_TX7N
◇OAM0_PCIE_RX7N
◇OAM0_PCIE_TX8P
◇OAM0_PCIE_RX8P
◇OAM0_PCIE_TX8N

◇OAM0_PCIE_RX8N
◇OAM0_PCIE_TX9P
◇OAM0_PCIE_RX9P
◇OAM0_PCIE_TX9N
◇OAM0_PCIE_RX9N
◇OAM0_PCIE_TX10P
◇OAM0_PCIE_RX10P
◇OAM0_PCIE_TX10N
◇OAM0_PCIE_RX10N
◇OAM0_PCIE_TX11P
◇OAM0_PCIE_RX11P
◇OAM0_PCIE_TX11N
◇OAM0_PCIE_RX11N
◇OAM0_PCIE_TX12P
◇OAM0_PCIE_RX12P
◇OAM0_PCIE_TX12N
◇OAM0_PCIE_RX12N
◇OAM0_PCIE_TX13P
◇OAM0_PCIE_RX13P
◇OAM0_PCIE_TX13N
◇OAM0_PCIE_RX13N
◇OAM0_PCIE_TX14P
◇OAM0_PCIE_RX14P
◇OAM0_PCIE_TX14N
◇OAM0_PCIE_RX14N
◇OAM0_PCIE_TX15P
◇OAM0_PCIE_RX15P
◇OAM0_PCIE_TX15N
◇OAM0_PCIE_RX15N
◇OAM1_PCIE_TX0P
◇OAM1_PCIE_RX0P
◇OAM1_PCIE_TX0N
◇OAM1_PCIE_RX0N
◇OAM1_PCIE_TX1P
◇OAM1_PCIE_RX1P

◇OAM1_PCIE_TX1N
◇OAM1_PCIE_RX1N
◇OAM1_PCIE_TX2P
◇OAM1_PCIE_RX2P
◇OAM1_PCIE_TX2N
◇OAM1_PCIE_RX2N
◇OAM1_PCIE_TX3P
◇OAM1_PCIE_RX3P
◇OAM1_PCIE_TX3N
◇OAM1_PCIE_RX3N
◇OAM1_PCIE_TX4P
◇OAM1_PCIE_RX4P
◇OAM1_PCIE_TX4N
◇OAM1_PCIE_RX4N
◇OAM1_PCIE_TX5P
◇OAM1_PCIE_RX5P
◇OAM1_PCIE_TX5N
◇OAM1_PCIE_RX5N
◇OAM1_PCIE_TX6P
◇OAM1_PCIE_RX6P
◇OAM1_PCIE_TX6N
◇OAM1_PCIE_RX6N
◇OAM1_PCIE_TX7P
◇OAM1_PCIE_RX7P
◇OAM1_PCIE_TX7N
◇OAM1_PCIE_RX7N
◇OAM1_PCIE_TX8P
◇OAM1_PCIE_RX8P
◇OAM1_PCIE_TX8N
◇OAM1_PCIE_RX8N
◇OAM1_PCIE_TX9P
◇OAM1_PCIE_RX9P
◇OAM1_PCIE_TX9N
◇OAM1_PCIE_RX9N
◇OAM1_PCIE_TX10P

◇OAM1_PCIE_RX10P
◇OAM1_PCIE_TX10N
◇OAM1_PCIE_RX10N
◇OAM1_PCIE_TX11P
◇OAM1_PCIE_RX11P
◇OAM1_PCIE_TX11N
◇OAM1_PCIE_RX11N
◇OAM1_PCIE_TX12P
◇OAM1_PCIE_RX12P
◇OAM1_PCIE_TX12N
◇OAM1_PCIE_RX12N
◇OAM1_PCIE_TX13P
◇OAM1_PCIE_RX13P
◇OAM1_PCIE_TX13N
◇OAM1_PCIE_RX13N
◇OAM1_PCIE_TX14P
◇OAM1_PCIE_RX14P
◇OAM1_PCIE_TX14N
◇OAM1_PCIE_RX14N
◇OAM1_PCIE_TX15P
◇OAM1_PCIE_RX15P
◇OAM1_PCIE_TX15N
◇OAM1_PCIE_RX15N
◇OAM2_PCIE_TX0P
◇OAM2_PCIE_RX0P
◇OAM2_PCIE_TX0N
◇OAM2_PCIE_RX0N
◇OAM2_PCIE_TX1P
◇OAM2_PCIE_RX1P
◇OAM2_PCIE_TX1N
◇OAM2_PCIE_RX1N
◇OAM2_PCIE_TX2P
◇OAM2_PCIE_RX2P
◇OAM2_PCIE_TX2N
◇OAM2_PCIE_RX2N

◇OAM2_PCIE_TX3P
◇OAM2_PCIE_RX3P
◇OAM2_PCIE_TX3N
◇OAM2_PCIE_RX3N
◇OAM2_PCIE_TX4P
◇OAM2_PCIE_RX4P
◇OAM2_PCIE_TX4N
◇OAM2_PCIE_RX4N
◇OAM2_PCIE_TX5P
◇OAM2_PCIE_RX5P
◇OAM2_PCIE_TX5N
◇OAM2_PCIE_RX5N
◇OAM2_PCIE_TX6P
◇OAM2_PCIE_RX6P
◇OAM2_PCIE_TX6N
◇OAM2_PCIE_RX6N
◇OAM2_PCIE_TX7P
◇OAM2_PCIE_RX7P
◇OAM2_PCIE_TX7N
◇OAM2_PCIE_RX7N
◇OAM2_PCIE_TX8P
◇OAM2_PCIE_RX8P
◇OAM2_PCIE_TX8N
◇OAM2_PCIE_RX8N
◇OAM2_PCIE_TX9P
◇OAM2_PCIE_RX9P
◇OAM2_PCIE_TX9N
◇OAM2_PCIE_RX9N
◇OAM2_PCIE_TX10P
◇OAM2_PCIE_RX10P
◇OAM2_PCIE_TX10N
◇OAM2_PCIE_RX10N
◇OAM2_PCIE_TX11P
◇OAM2_PCIE_RX11P
◇OAM2_PCIE_TX11N

◇OAM2_PCIE_RX11N
◇OAM2_PCIE_TX12P
◇OAM2_PCIE_RX12P
◇OAM2_PCIE_TX12N
◇OAM2_PCIE_RX12N
◇OAM2_PCIE_TX13P
◇OAM2_PCIE_RX13P
◇OAM2_PCIE_TX13N
◇OAM2_PCIE_RX13N
◇OAM2_PCIE_TX14P
◇OAM2_PCIE_RX14P
◇OAM2_PCIE_TX14N
◇OAM2_PCIE_RX14N
◇OAM2_PCIE_TX15P
◇OAM2_PCIE_RX15P
◇OAM2_PCIE_TX15N
◇OAM2_PCIE_RX15N
◇OAM3_PCIE_TX0P
◇OAM3_PCIE_RX0P
◇OAM3_PCIE_TX0N
◇OAM3_PCIE_RX0N
◇OAM3_PCIE_TX1P
◇OAM3_PCIE_RX1P
◇OAM3_PCIE_TX1N
◇OAM3_PCIE_RX1N
◇OAM3_PCIE_TX2P
◇OAM3_PCIE_RX2P
◇OAM3_PCIE_TX2N
◇OAM3_PCIE_RX2N
◇OAM3_PCIE_TX3P
◇OAM3_PCIE_RX3P
◇OAM3_PCIE_TX3N
◇OAM3_PCIE_RX3N
◇OAM3_PCIE_TX4P
◇OAM3_PCIE_RX4P

r2.0 UNUSABLE P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P
Pin names: OAM Pin map rev 1.0.xlsx – OCP generic, NOT AMD overlay
Documentation sheet – no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /02_Host_PCIe_Named_8Seat/
File: 02_host_pcie_stub.kicad_sch

Title: Host PE stub (all 8 seats via DNP PM8536 PRIMARY)

Size: A3 | Date: 2026-08-22 | Rev: Rev3_8Seat_PCBWay_v1
KiCad E.D.A. 9.0.9 | Id: 4/14

OAM0 connector instances — FIRST STUFF candidate (2 modules / 6 DNP OK).

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW0 PM8536B-FEI DNP PRIMARY (seats 0-3, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1-S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

P48V
P12V1
P12V2
P3V3
GND
OAM0_PVREF
OAM0_PE_REFCLKP
OAM0_PE_REFCLKN
OAM0_PERST#
OAM0_HOST_PWRGD
OAM0_MODULE_PWRGD
OAM0_WARMRST#
OAM0_PWRBRK#
OAM0_PRSNT0#
OAM0_SMBus_SLV_D
OAM0_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /03_OAM0_Connectors/
File: 03_oam0_connectors.kicad_sch

Title: OAM0 connectors

Size: A3	Date: 2026-08-22	Rev: Rev3_8Seat_PCBWay_v1
KiCad E.D.A. 9.0.9		Id: 5/14

OAM1 connector instances — FIRST STUFF candidate (2 modules / 6 DNP OK).

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW0 PM8536B-FEI DNP PRIMARY (seats 0–3, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1–S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

- P48V
- P12V1
- P12V2
- P3V3
- GND
- OAM1_PVREF
- OAM1_PE_REFCLKP
- OAM1_PE_REFCLKN
- OAM1_PERST#
- OAM1_HOST_PWRGD
- OAM1_MODULE_PWRGD
- OAM1_WARMRST#
- OAM1_PWRBRK#
- OAM1_PRSNT0#
- OAM1_SMBus_SLV_D
- OAM1_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /04_OAM1_Connectors/
File: 04_oam1_connectors.kicad_sch

Title: OAM1 connectors

Size: A3	Date: 2026-08-22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 6/14

OAM2 connector instances — electrically designed; module may be DNP.

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW0 PM8536B-FEI DNP PRIMARY (seats 0–3, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1–S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

- P48V
- P12V1
- P12V2
- P3V3
- GND
- OAM2_PVREF
- OAM2_PE_REFCLKP
- OAM2_PE_REFCLKN
- OAM2_PERST#
- OAM2_HOST_PWRGD
- OAM2_MODULE_PWRGD
- OAM2_WARMRST#
- OAM2_PWRBRK#
- OAM2_PRSNT0#
- OAM2_SMBus_SLV_D
- OAM2_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /05_OAM2_Connectors/
File: 05_oam2_connectors.kicad_sch

Title: OAM2 connectors

Size: A3	Date: 2026-08-22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 7/14

OAM3 connector instances — electrically designed; module may be DNP.

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW0 PM8536B-FEI DNP PRIMARY (seats 0-3, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1-S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

P48V
P12V1
P12V2
P3V3
GND
OAM3_PVREF
OAM3_PE_REFCLKP
OAM3_PE_REFCLKN
OAM3_PERST#
OAM3_HOST_PWRGD
OAM3_MODULE_PWRGD
OAM3_WARMRST#
OAM3_PWRBRK#
OAM3_PRSNT0#
OAM3_SMBus_SLV_D
OAM3_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /06_OAM3_Connectors/
File: 06_oam3_connectors.kicad_sch

Title: OAM3 connectors

Size: A3	Date: 2026-08-22	Rev: Rev3_8Seat_PCBWay_v1
KiCad E.D.A. 9.0.9		Id: 8/14

OAM4 connector instances — electrically designed; module may be DNP.

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW1 PM8536B-FEI DNP PRIMARY (seats 4-7, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1-S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

P48V
P12V1
P12V2
P3V3
GND
OAM4_PVREF
OAM4_PE_REFCLKP
OAM4_PE_REFCLKN
OAM4_PERST#
OAM4_HOST_PWRGD
OAM4_MODULE_PWRGD
OAM4_WARMRST#
OAM4_PWRBRK#
OAM4_PRSNT0#
OAM4_SMBus_SLV_D
OAM4_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /07_OAM4_Connectors/
File: 07_oam4_connectors.kicad_sch

Title: OAM4 connectors

Size: A3	Date: 2026-08-22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 9/14

OAM5 connector instances — electrically designed; module may be DNP.

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW1 PM8536B-FEI DNP PRIMARY (seats 4-7, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1-S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

P48V
P12V1
P12V2
P3V3
GND
OAM5_PVREF
OAM5_PE_REFCLKP
OAM5_PE_REFCLKN
OAM5_PERST#
OAM5_HOST_PWRGD
OAM5_MODULE_PWRGD
OAM5_WARMRST#
OAM5_PWRBRK#
OAM5_PRSNT0#
OAM5_SMBus_SLV_D
OAM5_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /08_OAM5_Connectors/
File: 08_oam5_connectors.kicad_sch

Title: OAM5 connectors

Size: A3	Date: 2026-08-22	Rev: Rev3_8Seat_PCBWay_v1
KiCad E.D.A. 9.0.9		Id: 10/14

OAM6 connector instances — electrically designed; module may be DNP.

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW1 PM8536B-FEI DNP PRIMARY (seats 4-7, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1-S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

P48V
P12V1
P12V2
P3V3
GND
OAM6_PVREF
OAM6_PE_REFCLKP
OAM6_PE_REFCLKN
OAM6_PERST#
OAM6_HOST_PWRGD
OAM6_MODULE_PWRGD
OAM6_WARMRST#
OAM6_PWRBRK#
OAM6_PRSNT0#
OAM6_SMBus_SLV_D
OAM6_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /09_OAM6_Connectors/
File: 09_oam6_connectors.kicad_sch

Title: OAM6 connectors

Size: A3 | Date: 2026-08-22 | **Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9 | Id: 11/14

OAM7 connector instances — electrically designed; module may be DNP.

Footprint: Molex 218910-1115 candidate (geometry). Hermaphroditic — mates with itself
(Farnell 2189101115: Mates With 2189101115; mated height 5.00 mm).

Pad nets assigned on the PCB from the v1.0 CSV for this seat:
P48V/P12V1/P3V3/GND, PE (16 lanes) toward SW1 PM8536B-FEI DNP PRIMARY (seats 4-7, x8/GCD),
PE_REFCLK / PERST# / HOST_PWRGD (ENABLE). S1-S7 have NO NET (no xGMI).
TEST*/RFU/DO_NOT_USE unmapped. PVREF is a module output — never drive.
P12V2 named from v1.0 but Unknown / may be NC. P48V is a LOCAL pour after a ~15 A fuse keepout.

MODULE_ID / LINK_CONFIG 1k pulldowns NOT placed (AMD overlay unknown).

P48V
P12V1
P12V2
P3V3
GND
OAM7_PVREF
OAM7_PE_REFCLKP
OAM7_PE_REFCLKN
OAM7_PERST#
OAM7_HOST_PWRGD
OAM7_MODULE_PWRGD
OAM7_WARMRST#
OAM7_PWRBRK#
OAM7_PRSNT0#
OAM7_SMBus_SLV_D
OAM7_SMBus_SLV_CLK

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /10_OAM7_Connectors/
File: 10_oam7_connectors.kicad_sch

Title: OAM7 connectors

Size: A3	Date: 2026-08-22	Rev: Rev3_8Seat_PCBWay_v1
KiCad E.D.A. 9.0.9		Id: 12/14

TWO PM8536B-FEI KEEPOUTS — PRIMARY, DNP. No fake schematic pins.

U_SW0 PM8536B-FEI DNP seats 0-3   16 US + 64 DS   x8 per GCD
U_SW1 PM8536B-FEI DNP seats 4-7   16 US + 64 DS   x8 per GCD

Package (Verified Microchip PFX table): 96-lane Gen3, 1311-ball 37.5 x 37.5 mm FCBGA,
1.0 mm pitch. Street ~$460-475, ~18 wk. PCBWay can assemble 1.0 mm without HDI;
plan 12-16 layers (not 4). 12L 2.0 mm is the stuffed-switch TARGET.
8L 2.0 mm is a cheaper DNP-switch / mezz+power option only.

Do NOT populate until Eli buys the parts. Courtyard/keepout only — a DNP box, no fake pins.

Cheaper 80-lane alt in docs only (not placed): PEX8780-AB80BI G, 35 mm 1156-FCBGA.
2-seat alt in docs only (not placed): PM8533B-F3EI (48-lane, 27 mm, 1.0 mm).

Each MI250X GCD has its own PCIe Gen4 x16 (AMD + Hot Chips 34). Host trains Gen3.
OAM v1.5: one PE x16 on Conn0; a second host x16 may be SERDES_7 on Conn1.
GCD1 = SERDES_7 is Inferred — NO public AMD overlay proving it. Do NOT route S1-S7.

Host X11DPH-T (NOT on this PCB): 3x Gen3 x16 + 4x Gen3 x8 = 80 Gen3 lanes.
Two CPU x16 become the two switch uplinks (silk + connector keepout, MPN Unknown).
Do not invent CEM / SlimSAS / MCIO / retimer / xGMI.

8 x 2 x x16 = 256 DS vs 80 host — x16-per-GCD is impossible. x8 per GCD is the plan.

Refclk / PERST# / HOST_PWRGD per seat per OAM v1.5
(HOST_PWRGD >= 100 ms after MODULE_PWRGD). AMD delays still Unknown.

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed-switch TARGET. 2x PM8536 DNP. Star-fed P4
Pin names: OAM Pin map rev 1.0.xlsx — OCP generic, NOT AMD overlay
Documentation sheet — no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026-08-18; 1.2A/contact OPEN.

Sheet: /11_PM8536_DNP/
File: 11_pcie_switch_unknown.kicad_sch

Title: PM8536B-FEI DNP x2 PRIMARY (keepout only)

Size: A3	Date: 2026-08-22	**Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9		Id: 13/14

Pads explicitly UNMAPPED on every seat (no net, no copper).
Named in the v1.0 generic map but reserved, do-not-use, or TEST pins
whose AMD MI250X function is Unknown. Do not invent pullups or straps.
S1–S7 / QSFP sideband also have no net (no xGMI). All 8 seats are otherwise electrically named.

Conn0 C4 TEST_MODE#
Conn0 F1 RFU
Conn0 F2 RFU
Conn0 F26 TEST0
Conn0 F29 TEST1
Conn0 F30 TEST2
Conn0 F33 TEST3
Conn0 F34 TEST4
Conn0 F37 TEST5
Conn0 F38 TEST6
Conn0 F41 TEST7
Conn0 F42 TEST8
Conn0 F49 TEST9
Conn0 F5 RFU
Conn0 F50 RFU
Conn0 H58 DO_NOT_USE
Conn0 J60 DO_NOT_USE
Conn0 K58 DO_NOT_USE
Conn0 L60 DO_NOT_USE
Conn1 A7 RFU
Conn1 A8 RFU
Conn1 B5 RFU
Conn1 B6 RFU
Conn1 C3 RFU
Conn1 C4 RFU
Conn1 C59 RFU
Conn1 C60 RFU
Conn1 C7 RFU
Conn1 C8 RFU
Conn1 E31 RFU
Conn1 E32 RFU
Conn1 E39 RFU
Conn1 E40 RFU
Conn1 F13 TEST10
Conn1 F14 TEST11
Conn1 F17 TEST12
Conn1 F18 TEST13
Conn1 F21 TEST14
Conn1 F22 RFU
Conn1 F29 RFU
Conn1 F30 RFU
Conn1 F33 RFU
Conn1 F34 RFU
Conn1 F37 RFU
Conn1 F38 RFU
Conn1 F41 RFU
Conn1 F42 RFU
Conn1 F45 RFU
Conn1 G3 RFU
Conn1 G31 RFU
Conn1 G32 RFU
Conn1 G4 RFU
Conn1 G7 RFU
Conn1 G8 RFU
Conn1 H5 RFU
Conn1 H6 RFU
Conn1 J3 RFU
Conn1 J4 RFU
Conn1 J64 RFU

r2.0 UNUSABLE. P3V3=Conn0 C1/C2. 12L 2.0mm stuffed–switch TARGET. 2x PM8536 DNP. Star–fed P4
Pin names: OAM Pin map rev 1.0.xlsx – OCP generic, NOT AMD overlay
Documentation sheet – no invented nets
DO NOT FABRICATE. DO NOT ENERGIZE P48V. Molex 60V written 2026–08–18; 1.2A/contact OPEN.

Sheet: /12_Unmapped_AMD/
File: 12_unmapped_amd.kicad_sch

Title: Unmapped / AMD–unknown pads

Size: A3 | Date: 2026–08–22 | **Rev: Rev3_8Seat_PCBWay_v1**
KiCad E.D.A. 9.0.9 | Id: 14/14